

project summary
phase I

Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

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Fuel-spray atomization and evaporation sit at the heart of advanced combustion systems. In this proposal, we develop AI-enhanced numerical methods for “digital twin” atomization experiments designed to improve the efficiency and performance of jet engines and rotating detonation rocket engines.

What makes these problems so difficult is their multiscale character. Large liquid structures break into sheets and ligaments, then into droplets and mist, while evaporation continuously converts liquid fuel into a gaseous phase ready to react with oxidizer. Meanwhile, breakup, evaporation, and heat transfer reshape the surrounding turbulent flow, which then feeds back on the atomization process itself. In rotating detonation engines, the picture is even more delicate: some droplets survive the initial atomization stage and are later shattered by the detonation wave into a fine fog, after which reactions help drive the next cycle. The engine works best only within a narrow window, where the remaining pre-shock droplets are large enough to survive long enough to matter, but not so large that they reduce efficiency.

The difficulty, of course, is that the highest-fidelity simulations and the most informative experiments are both expensive. Our goal is therefore not to replace physics with AI, but to use AI to make better use of scarce high-value data within a conservative, interface-resolved simulation framework. The proposed work has three tightly connected parts: (1) data assimilation, (2) AI learning of multiphase-flow sub-scale corrections, and (3) verification and validation. In the first part, we will develop a compressible multiphase nudging capability that assimilates interface, velocity, and temperature information from high-fidelity simulations and/or experiments into a coarse-grid solver while preserving volume and mass. This step is essential because it provides a physically consistent way to align coarse-grid solutions with reference dynamics, rather than relying on raw coarse/fine mismatches that may be dominated by trajectory drift.

In the second part, the correction generated by this nudging-based alignment will be used to train a neural-network model to learn the unresolved interfacial and near-interface physics missing from underresolved simulations. In practice, the AI model will be trained on one-step correction terms extracted from an interface-resolved solver, using coarse-grid flow variables together with reconstructed interface features such as volume fraction, centroids, interface normals, and curvature. In this way, the learned model remains embedded within the governing equations of the underlying multiphase solver, where it acts as a localized, physics-informed correction rather than as a black-box surrogate. In the third part, we will test these learned corrections on atomization and spray problems outside the original training set, comparing coarse-grid results obtained with and without the AI model and quantifying the resulting improvement in interface prediction. The overall aim is to create a practical AI-enabled pathway for extending the predictive reach of high-fidelity multiphase simulation while preserving the physical structure and credibility of the underlying numerical method.